ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–13) | Class 9 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(d)** a feather	Air resistance matters for feathers.
2	**(a)** 20 per cent	30/150.
3	**(c)** 10 g	Conservation of mass.
4	**(d)** 16	Sum of electrons in a neutral atom.
5	**(b)** 12 m/s	v = at.
6	**(a)** 6 kg	m = F/a.
7	**(c)** zero	cos 90 = 0.
8	**(b)** 0.68 m	v/f.
9	**(d)** hypertonic	Water leaves.
10	**(c)** increase in girth	Secondary growth.
11	**(d)** Protista	Euglena.
12	**(b)** vegetative propagation by suckers	Banana is seedless.
13	**(b)** 500 s	d/v.
14	**(d)** slowly stops swinging	Friction at the pivot and air resistance.
15	**(d)** fractional distillation	Boiling-point difference.
16	**(c)** $Al_2(SO_4)_3$	Al3+ and SO4 2-.
17	**(b)** 12	2, 8, 2.
18	**(d)** 5 m/s	100/20.
19	**(d)** 200 N	600/3.
20	**(a)** 9	0.06 x 150.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Acceptable: air hockey puck on a smooth table (1); Fails: car braking on a rough road (1).	1+1
22	8/100 x 500 (1); = 40 g (1).	1+1
23	Mass of water = 18 g (1); Reactants = 18 g, products = 18 g (1).	1+1
24	A = 24 (1); 2, 8, 2 (1).	1+1
25	s = 15 x 20 = 300 m (1); Horizontal line; area gives distance (1).	1+1
26	Net force 12 N (1); a = 2.4 m/s squared (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	T = 32/20 = 1.6 s (1); Reduces reaction-time error (1); Gives better precision (1).	1+1+1
28	36/200 x 100 = 18 per cent (1); New solution = 100 g (1); New percentage = 36 per cent (1).	1+1+1
29	C = 3/11 x 55 = 15 g (1); O = 8/11 x 55 = 40 g (1); Total = 55 g (1).	1+1+1
30	A = 40 (1); 2, 8, 8, 2; valency 2 (1); Calcium (1).	1+1+1
31	a = 2 m/s squared (1); s = 25 x 5 = 125 m (1); Average velocity 25 m/s (1).	1+1+1
	OR: a = 5 m/s squared (1); v = 20 m/s (1); s = 40 m (1).	1+1+1
32	Start: KE 225 J (1); At 5 m: PE 100 J, KE 125 J (1); Top: PE 225 J (11.25 m) (1).	1+1+1
	OR: f = 400 Hz (1); T = 0.0025 s (1); lambda = 0.85 m (1).	1+1+1

SECTION — D (Long Answer) — Answers — 6 × 5 = 30

Q.	Expected Answer / Value Points	Marks

33	Periods: 1.0, 1.42, 1.74, 2.0 s (1); T increases with length (1); Increase is not proportional (1); No friction or air resistance assumed (1); Small amplitude assumed (1).	1+1+1+1+1
34	A = 5 per cent (1); B = 10 per cent (1); C = 15 per cent (1); Mixing equal masses of A and C gives 10 per cent (1); Ranking C > B > A (1).	1+1+1+1+1
35	Ratio H : O in the first sample = 1 : 8 (1); Ratio in the second sample = 2 : 16 = 1 : 8 (1); The ratio is the same for different samples (1); Law: a pure compound always contains the same elements in a fixed ratio by mass (1); Proust gave this law; it distinguishes a compound from a mixture (1).	1+1+1+1+1
36	p = e = 13, n = 14 (1); 2, 8, 3 (1); Valency 3 (1); Al^{3+} (1); Al_2O_3 (1).	1+1+1+1+1
	OR: Maximum velocity 20 m/s (1); Retardation 4 m/s squared (1); Distances 100, 400 and 50 m (1); Total 550 m (1); Trapezium graph (1).	1+1+1+1+1
37	a = 0.5 m/s squared (1); Retardation 1.5 m/s squared (1); Time = 20 s (1); Distance = 900/3 = 300 m (1); Second law used (1).	1+1+1+1+1
	OR: Start: PE 900 J (1); After 1 s: fallen 5 m, KE 100 J, PE 800 J (1); After 2 s: fallen 20 m, KE 400 J, PE 500 J (1); Ground: KE 900 J (v = 30 m/s) (1); Total 900 J (1).	1+1+1+1+1
38	Nearer cliff = 170 m (1); Farther cliff = 340 m (1); Distance between cliffs = 510 m (1); At the centre: 255 m each (1); Time = 510/340 = 1.5 s for both echoes (1).	1+1+1+1+1
	OR: Double membrane with pores (1); Nucleolus: ribosome synthesis (1); Chromatin: DNA and proteins (1); Control of cell activities (1); Heredity and division (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–13) | Class 9 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(d)** greater for the 256 Hz fork	lambda proportional to 1/f.
2	**(c)** 8	Same number.
3	**(c)** squamous epithelium	Alveoli.
4	**(c)** gills throughout life	Fish breathe with gills.
5	**(a)** bisexual flower	Has both.
6	**(d)** coastal low-lying areas	Flooding.
7	**(c)** 2500	1 g/cm3 = 1000 kg/m3.
8	**(c)** sublimation, dissolving and filtering, evaporation	Camphor first.
9	**(a)** Mg and Cl	Metal and non-metal.
10	**(d)** 35.5 u	26.25 + 9.25.
11	**(a)** 30 m/s	t up = 3 s; u = gt.
12	**(d)** 12 N	Equal to the applied force at the verge of motion.
13	**(c)** 4	Effort arm/load arm.
14	**(c)** quality or timbre	Waveform.
15	**(b)** mitochondria	Energy.
16	**(a)** hinge joint	One plane.
17	**(c)** Animalia, Chordata, Mammalia, Carnivora, Felidae, Panthera, tigris	Hierarchy.
18	**(a)** 46	23 + 23.
19	**(a)** evaporation	Heating.
20	**(c)** 8 m/s	100/12.5.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	W = 200 J (1); KE = 200 J (1).	1+1
22	v = 340 m/s (1); T = 0.05 s; 10 oscillations 0.5 s (1).	1+1
23	RBC has no cell wall (1); Plant cell wall resists the expansion (1).	1+1
24	They are actively dividing (1); Dense cytoplasm and thin walls (1).	1+1
25	Protista (1); Unicellular eukaryote (1).	1+1
26	They come from a single parent (1); Genetically identical clones (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	In water: mass increases (1); In salt solution: mass decreases (1); Osmosis through membranes (1).	1+1+1
28	Water enters, guard cells swell (1); Stoma opens (1); Water leaves, they become flaccid and the stoma closes (1).	1+1+1
29	Banyan: dicot angiosperm; rice: monocot (1); Fern: pteridophyte; moss: bryophyte (1); Spirogyra: alga; pine: gymnosperm (1).	1+1+1
30	Sperm: small, motile, tail (1); Ovum: large, non-motile, stored food (1); Sperm: DNA delivery; ovum: nutrition for zygote (1).	1+1+1
31	Snow reflects 85 per cent (1); Forest and ocean reflect little, absorb a lot (1); Melting of snow increases warming (1).	1+1+1
	OR: Atom about 10^-10 m (1); Sun about 1.5 x 10^11 m (1); Easy comparison (1).	1+1+1
32	KNO3 (1); Its solubility falls sharply on cooling (1); NaCl solubility changes little (1).	1+1+1

	OR: NaCl ionic, CCl_4 covalent (1); NaCl high melting, CCl_4 low (1); NaCl conducts in solution, CCl_4 does not (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Thin section, stain and mount (1); Epidermis: protection (1); Cortex: storage and support (1); Vascular bundles: conduction (1); Pith: storage (1).	1+1+1+1+1
34	Mushroom: Fungi; bacterium: Monera (1); Euglena: Protista (1); Rose: angiosperm; fern: pteridophyte; moss: bryophyte (1); Whale: mammal; snake: reptile; pigeon: bird (1); Earthworm: annelid (1).	1+1+1+1+1
35	Gametes formed in testes and ovaries (1); Fertilisation in the oviduct (1); Implantation in the uterus (1); Development with placenta (1); Birth after about 9 months (1).	1+1+1+1+1
36	Energy received = 2 kWh (1); Snow absorbs 0.3 kWh (1); Forest absorbs 1.8 kWh (1); Difference 1.5 kWh (1); Explains why forests are warmer (1).	1+1+1+1+1
	OR: Initial observation (1); Hypothesis (1); Repeated experiments (1); Peer review (1); Wide acceptance (1).	1+1+1+1+1
37	Air filtered, CO2 and water removed (1); Compressed and cooled (1); Liquid air fractionated (1); Nitrogen, argon, oxygen collected (1); Uses of each gas (1).	1+1+1+1+1
	OR: O = 1.5 g (1); Cu : O = 6 : 1.5 = 4 : 1 (1); CuO ratio = 63.5 : 16 = 3.97 : 1 (1); Agrees within error (1); Compound confirmed as CuO (1).	1+1+1+1+1
38	C-12: 6 p, 6 n (1); C-13: 7 n (1); C-14: 8 n, radioactive (1); Dating of fossils by decay of C-14 (1); Half life about 5730 years (1).	1+1+1+1+1
	OR: Speed 20 m/s (1); Reaction distance 10 m (1); Braking distance 40 m (1); Total 50 m (1); Just stops at the obstacle (1).	1+1+1+1+1

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–13) | Class 9 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** 10 g	4/100 x 250.
2	**(b)** 28 g	0.8235 x 34.
3	**(a)** 1	2, 8, 8, 1.
4	**(a)** 100 m	s = (1/2)at squared.
5	**(d)** 6 N	Change in momentum/time.
6	**(a)** 20 m/s	sqrt(2gh).
7	**(c)** 20	3.4/0.17.
8	**(c)** growing region	Meristem.
9	**(a)** vascular cambium	Rings.
10	**(c)** monocot	Monocot.
11	**(a)** absent	Pregnancy.
12	**(c)** nitrogen fixation	Bacteria and lightning.
13	**(b)** supports the law	Agreement supports it.
14	**(a)** sublimation	Iodine sublimes.
15	**(b)** $(NH_4)_3PO_4$	NH4+ and PO4 3-.
16	**(b)** 4	2, 8, 4.
17	**(b)** 48 km/h	Harmonic mean 2 x 40 x 60/100.
18	**(c)** 60 kg	Mass is constant.
19	**(a)** 120 J	Only vertical displacement counts.
20	**(a)** 0.00588 s	lambda/v.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	t = 500 s (1); = 8.3 minutes (1).	1+1
22	Easy to read and compare (1); 3 x 10^8 (1).	1+1
23	To measure the temperature of the vapour (1); Which equals the boiling point of the liquid (1).	1+1
24	H has valency 1 and O has valency 2 (1); Two H atoms are needed for one O atom: H_2O (1).	1+1
25	(63 x 70 + 65 x 30)/100 (1); = 63.6 u (1).	1+1
26	t = 2 s (1); v = 20 m/s (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	2, 8, 3 (1); Z = 13 (1); Valency 3 (1).	1+1+1
28	s_A = t squared; s_B = 20t (1); Equal when t = 20 s (1); Distance 400 m (1).	1+1+1
29	Change in momentum 50000 (1); F = 100000 N (1); Weight = 50000 N; force is twice the weight (1).	1+1+1
30	At 50 cm: 40 N, MA 2.5 (1); At 1 m: 20 N, MA 5 (1); Longer effort arm gives larger MA (1).	1+1+1
31	t = d/v (1); 3400/340 (1); 10 s (1).	1+1+1
	OR: Bacterium has no nucleus, only a nucleoid (1); Animal cell has a nucleus with chromosomes (1); Bacteria have cell walls (1).	1+1+1
32	Plants: xylem and phloem (1); Animals: blood and lymph (1); Plants have meristems; animals have dividing cells everywhere (1).	1+1+1
	OR: Aves (1); Hollow bones for flight (1); Warm-blooded (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Surface irregularities (1); Depends on nature and normal force (1); f = 60 N (1); Net 20 N (1); a = 1 m/s squared (1).	1+1+1+1+1
34	Efficiency = useful output / input x 100 (1); = 600/800 x 100 (1); = 75 per cent (1); Energy lost = 800 - 600 (1); = 200 J, wasted mainly as heat by friction (1).	1+1+1+1+1
35	A is taller (1); B has more waves in the same time (1); A louder (1); B higher pitch (1); Wavelengths: B is half of A (1).	1+1+1+1+1
36	Nucleus: manager (1); Ribosomes: workers (1); ER and Golgi: transport and packing (1); Mitochondria: power (1); Lysosomes: waste disposal; membrane: gate (1).	1+1+1+1+1
	OR: Aquatic: aerenchyma (1); Desert: thick cuticle, sunken stomata (1); Land: strong xylem (1); Root tissues (1); Examples (1).	1+1+1+1+1
37	Birds: feathers, hollow bones, wings (1); Streamlined body (1); Fish: fins, gills (1); Swim bladder, scales (1); Streamlined (1).	1+1+1+1+1
	OR: Seed coat (1); Cotyledons (1); Embryo: radicle, plumule (1); Endosperm in maize (1); Germination steps (1).	1+1+1+1+1
38	Forest absorbs 600 tonnes (1); Cars emit 1440 tonnes (1); Forest offsets less than half (1); Need for emission cuts (1); Plant more trees (1).	1+1+1+1+1
	OR: Air breathed in a day (1); Heartbeats in a life (1); Mass of water in a school tank (1); Assumed average values (1); Uncertainty from approximations (1).	1+1+1+1+1

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